STANES SCHOOL

1044, Avinashi Road, Coimbatore – 18.

UNIT TEST – 2 2026-2027

Class : XII **Duration : 2 Hours**

Subject : Informatics Practices (065) **Max. Marks : 50**

Section A (14 x 1 = 14 Marks)

1. Which of the following SQL functions is used to find the position of a substring in a string?

a) LENGTH() b) INSTR() c) SUBSTRING() d) TRIM()

2. In a network topology where every node is directly connected to every other node, the topology is known as:

a) Star b) Bus c) Mesh d) Tree

3. Which SQL function is used to display the name of the day (e.g., Monday, Tuesday) from a date?

a) DAY() b) DATE() c) DAYNAME() d) MONTHNAME()

4. Which of the following is a network device used to connect two different networks and forward data packets between them?

a) Hub b) Switch c) Router d) Repeater

5. Which SQL function is used to remove leading and trailing spaces from a string?

a) LTRIM() b) RTRIM() c) TRIM() d) LENGTH()

6. What will be the output of the following SQL query?

SELECT ROUND(15.78, 0);

a) 15 b) 16 c) 15.8 d) 15.78

7. Which aggregate function is used to count the total number of non-NULL values in a column?

a) COUNT(*) b) COUNT(column_name)

c) SUM(column_name) d) AVG(column_name)

8. Which type of network is formed when the Hyderabad campus of a company connects to its Bengaluru headquarters?

a) PAN b) LAN c) MAN d) WAN

9. Which of the following SQL commands is used to sort rows in ascending or descending order of values in a specific column?

a) GROUP BY b) ORDER BY c) SORT BY d) HAVING

10. Which SQL function will return the remainder when one number is divided by another?

a) MOD() b) ROUND() c) POWER() d) TRIM()

11. Which of the following SQL functions will convert the string 'hello' to 'HELLO'?

a) LCASE('hello') b) LOWER('hello') c) UCASE('hello') d) SUBSTRING('hello')

12. A table has 3 candidate keys. Among these, one is chosen as the primary key. How many alternate keys does the table have?

a) 1 b) 2 c) 3 d) 4

13. Which SQL clause is used to filter groups created by the GROUP BY clause?

a) WHERE b) ORDER BY c) HAVING d) DISTINCT

14. What is the purpose of a modem in a computer network?

a) To amplify signals b) To convert digital signals to analog signals and vice versa

c) To route data packets between networks d) To connect devices within a LAN

Section B (4 x 2 = 8 Marks)

15. A. Differentiate between ORDER BY and GROUP BY in SQL with suitable examples.

OR

B. Explain the concepts of PAN and LAN with a suitable example for each.

16. Write the output of the following SQL queries based on the table **PRODUCT**:

Prod_ID	Prod_Name	Price	Qty
P101	Laptop	60000	15
P102	Mouse	500	50
P103	Keyboard	800	30
P104	Monitor	12000	10
P105	Printer	15000	5

I. SELECT MAX(Price), MIN(Price) FROM PRODUCT;

II. SELECT COUNT(*) FROM PRODUCT WHERE Qty > 20;

17. A. Write a SQL query to display the current date and the month name of the current date.

OR

B. Explain the difference between a hub and a switch in a computer network.

18. Consider the following table **EMPLOYEE**:

Emp_ID	Emp_Name	Dept	Salary
E01	Raj	IT	75000
E02	Priya	HR	55000
E03	Amit	IT	80000
E04	Sneha	Finance	65000

Write a SQL query to display the department-wise average salary, but only for those departments where the average salary is greater than 60000.

Section C (2 x 3 = 6 Marks)

19. Differentiate between Star topology and Bus topology with suitable diagrams & explanations.

20. Write SQL queries for the following:

I. To display the position of the substring 'Science' in the string 'Computer Science'.

II. To display the length of the string 'Informatics Practices'.

III. To extract the first 5 characters from the string 'Database Management'.

Section D (3 x 4 = 12 Marks)

21. Consider the table EMPLOYEE and DEPARTMENT given below:

Table: EMPLOYEE

Emp_ID	Emp_Name	Dept_ID	Salary	Join_Date
101	Arvind	D01	75000	2020-05-15
102	Bhavna	D02	55000	2019-08-20
103	Chetan	D03	65000	2021-01-10
104	Divya	D01	80000	2018-11-25
105	Esha	D04	60000	2022-03-01
106	Farhan	D01	90000	2020-07-12
107	Gauri	D02	48000	2021-09-05

Table: DEPARTMENT

Dept_ID	Dept_Name	Location
D01	IT	Mumbai
D02	HR	Delhi
D03	Finance	Mumbai
D04	Marketing	Chennai
D05	Sales	Bangalore

Write SQL queries for the following:

I. To display the employee name, department name, and salary of all employees using an equi-join.

II. To display the department name and the total salary paid to each department using natural join.

III. To display the employee name, department name, and joining month for employees working in the 'IT' department.

IV. To display the department name, location, and the number of employees in each department, but only for departments that have more than 2 employees.

22. XYZ Tech Solutions is setting up its new office in Hyderabad with four departments: HR, Finance, IT, and Logistics. The distances between departments and the number of computers in each department are given below:

Distances between departments:

From	To	Distance (in meters)
HR	Finance	50
HR	IT	175
HR	Logistics	90
Finance	IT	60
Finance	Logistics	70
IT	Logistics	60

Number of Computers in each department:

Department	Number of Computers
HR	60
Finance	40
IT	90
Logistics	30

I. Suggest the most suitable department to install the server. Justify your reason.

II. Draw a suitable cable layout of connections between the departments in the campus.

III. Which hardware device will you suggest to connect all the computers within each department?

IV. Which network topology is most suitable for connecting the departments in this campus? Justify your answer.

23. Define the following network devices and explain their functions with block diagrams:

I. Repeater

II. Router

III. Gateway

IV. Switch

Section E (2 x 5 = 10 Marks)

24. Consider the table STUDENT and MARKS given below:

Table: STUDENT

StudentID	Name	Class	City
101	Ankit Sharma	12	Delhi
102	Priya Singh	11	Mumbai
103	Rohan Verma	12	Delhi
104	Shreya Patel	11	Chennai
105	Rehan Khan	12	Mumbai

Table: MARKS

StudentID	Subject	Score
101	Mathematics	85
101	Physics	78
102	Chemistry	88
103	Mathematics	92
104	Biology	81
105	Computer Science	93
102	Mathematics	76

Predict the output of the following SQL queries:

I. SELECT Name FROM STUDENT WHERE Class = '12' ORDER BY Name ASC;

II. SELECT DISTINCT UCASE(Subject) FROM MARKS WHERE Score > 80;

III. SELECT STUDENT.Name, MARKS.Subject, MARKS.Score FROM STUDENT, MARKS WHERE STUDENT.StudentID = MARKS.StudentID;

IV. SELECT STUDENT.Name, SUM(MARKS.Score) AS Total_Marks FROM STUDENT, MARKS WHERE STUDENT.StudentID = MARKS.StudentID GROUP BY STUDENT.Name HAVING COUNT(MARKS.Subject) > 1;

V. SELECT Subject, AVG(Score) AS Average_Score FROM MARKS GROUP BY Subject;

25. Shiva Corporation is planning a new campus in Pune while maintaining its headquarters in Mumbai. The campus will have four buildings: Administration, Sales, Development, and Support. As a network expert, you are tasked with proposing the best network solutions for their needs based on the following:

Distances between buildings:

From	To	Distance (in meters)
Administration	Sales	60
Administration	Development	90
Administration	Support	120
Sales	Development	50
Sales	Support	70
Development	Support	45

Number of Computers in each building:

Building	Number of Computers
Administration	120
Sales	40
Development	70
Support	25
Mumbai HQ	150

I. Suggest the most suitable building in the Pune campus to install the server. Justify your reason.

II. Suggest and draw a cable layout of connections between the buildings inside the Pune campus.

III. Which hardware device will you suggest to connect all the computers within each building?

IV. The organization plans to provide a high-speed wired link with its Mumbai headquarters. Which type of cable will be most suitable for this job?

V. Which type of network (PAN, LAN, MAN, or WAN) will be formed when connecting the Pune campus to the Mumbai headquarters?